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/*****
*****
*检验1996-2013年的全体A股三因子回归模型
*参考文献: the cross-section of expected stock returns(1992), Fama & French
*****
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/*建立逻辑库 并放入股票月收益monthly,财务数据assetdebt,profit及金融行业公司代码
finastkcd */

%let path=G:\empiricalfinance\Fama_French1992;
libname fmreg "&path.\dataset";
filename odsout "&path.\result";
ods html path=odsout body="Fama_French_1992.htm"
(title="Fama_French_1992");

data a1;
set fmreg.assetdebt;
where stkcd like '60%' or stkcd like '00%' or stkcd like '300%'; /*选出所
有上海和深圳A股*/
year=input(substr(left(Accper),1,4),4.); /*提取年度*/
month=input(substr(left(Accper),6,2),2.); /*提取月度*/
if year>=1996; /*从1996年开始*/
if month=12; /*选择12月月末的数据（即年报值）*/
rename A003000000=equity A001000000=asset;
run;
proc sort data=a1;
by stkcd year;
run; /*数据集a1, 包含符合条件个股的年报里的股东权益和总资产信息*/

data a2;
set fmreg.profit;
where stkcd like '60%' or stkcd like '00%' or stkcd like '300%'; /*选出所
有上海和深圳A股*/
year=input(substr(left(Accper),1,4),4.); /*提取年度*/
month=input(substr(left(Accper),6,2),2.); /*提取月度*/
if year>=1996; /*从1996年开始*/
if month=12; /*选择12月月末的数据（即年报值）*/
rename B002000000=netprofit B001101000=revenue;
run;
proc sort data=a2;
by stkcd year;
run; /*数据集a2, 包含符合条件个股的年报里的净利润和营业收入信息*/

data fmreg.fmreg;
merge a1 a2 fmreg.finastkcd(in=id);/*数据集finastkcd, 每年金融行业公司的代码
*/
by stkcd year;
if id=0; /*剔除金融行业公司*/
run; /*数据集fmreg.fmreg, 合并所有财务数据（除金融行业公司）*/

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/*****步骤1：数据预处理*****/
/*****1.1月度回报数据集*****/
data monre1;
set fmreg.monthly;
where stkcd like '60%' or stkcd like '00%' or stkcd like '300%'; /*选出所
有上海和深圳A股*/
year=input(substr(left(Trdmnt),1,4),4.); /*提取年度*/
month=input(substr(left(Trdmnt),6,2),2.); /*提取月度*/
rename Msmvttl=marketvalue Mretwd=return Mclsprc=clprice;
keep stkcd year month Msmvttl Mretwd Mclsprc;
run;
data monre2; set monre1;
if return^=.; /*数据集fmreg.monthly中，提取出无缺省值的信息*/
marketvalue = marketvalue * 1000; /*国泰安数据库的总市值数据单位：千元*/
run;
proc sort data=monre2; by stkcd year month; run;
/*数据集monre2，包含符合条件个股的月度收益率、月末收盘价和月末总股本数据*/

/*填充有实际值首尾月份之间的缺省月度的资本收益率（首为其第一次收益率有值日；尾统一为
2013年12月）*/
data monre_everym1;
set monre2(keep=stkcd year month);
by stkcd year month;
if first.stkcd;
run; /*提取每只股票第一个交易月度*/
data monre_everym2; set monre_everym1;
endyear=2013;
endmonth=12;
run; /*统一最后一个月度为2013年12月*/
data monre_everym3;
merge monre_everym1 monre_everym2;
by stkcd;
run; /*数据集monre_everym3，包含每只股票的第一个交易月度和最后一个交易月度*/
data monre_everym4; set monre_everym3;
monthgap = (endyear-year-1)*12 + (12-month) + endmonth + 1; /*从开始到结束
的总月数计算*/
drop endyear endmonth;
run; /*数据集monre_everym4，包含每只股票的第一个交易月度和持续交易月度数（以日历计
算）*/
data monre_everym5; set monre_everym4;
do i=1 to monthgap;
    output;
    if month=12 then do; month=1; year=year+1; end;
    else month=month+1;
end;
drop monthgap i;
run; /*数据集monre_everym5，包含每只股票的持续交易月度（以日历计算）*/
data monre_everym6;
merge monre2 monre_everym5;
by stkcd year month;
run; /*数据集monre_everym6，包含完整月度的资本收益率（停牌月度为缺省值）*/

/*****1.2财务因子数据集*****/

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%macro ratiofactor(infile=,var=,factor=);
    data finance1; set &infile; keep stkcd year &var; run;
    data finance2; set finance1; where &var^=.; run;
    data proxy1;
    set monre2(drop=return);
    where month=12; /*选择12月月末的数据（即用12月末的总市值计算指标）*/
    drop month;
    run;

    data proxy2;
    merge finance2(in=id) proxy1;
    by stkcd year;
    if id;
    &factor = &var / marketvalue;
    if &factor ^= .; /*计算指标，且指标不能缺省（缺省值来自于12月市值缺省）*/
    keep stkcd year &factor;
    run; /*数据集proxy2，包含计算的指标最原始数据*/

    data proxy3; set proxy2;
    where &factor > 0;
    run; /*数据集proxy4，仅提取正的指标*/
    proc sort data=proxy2 out=&factor._all; by year stkcd; run;
    proc sort data=proxy3 out=&factor; by year stkcd; run;
    /*若仅需要正值，则输出数据集为&factor; 若需要全值，则输出数据集为&factor._all*/
%mend;

/*计算指标: BM*/
%ratiofactor(infile=fmreg.fmreg,var=equity,factor=bm);
/*计算指标: AM*/
%ratiofactor(infile=fmreg.fmreg,var=asset,factor=am);
/*计算指标: RM*/
%ratiofactor(infile=fmreg.fmreg,var=revenue,factor=rm);
/*计算指标: EP_positive*/
%ratiofactor(infile=fmreg.fmreg,var=netprofit,factor=ep);
data ep_positive; set ep_all;
if ep < 0 then ep_positive = 0; else ep_positive = ep;
drop ep;
run;
/*计算指标: EP_negative*/
data ep_negative; set ep_all;
if ep < 0 then ep_negative = 1; else ep_negative = 0;
drop ep;
run;
/*计算指标: AB*/
data ab;
merge am bm;
by year stkcd;
ab = am / bm; /*总资产比净资产 = 总资产市值比 / 净资产市值比*/
if ab=. then delete; /*AB缺省情况均来自于净资产值缺省（负数）情况，因此删除*/
drop am bm;
run;

/*****1.3贝塔因子数据集*****/

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/*1.3.1全部A股的总市值加权指数计算*/
data vwindex1; set monre_everym6; by stkcd year month;
marketvalue_1 = lag(marketvalue); /*上一个月的总市值*/
if first.stkcd then marketvalue_1 = .; /*每只股票头一个月的上一个月总市值为缺省值*/
label marketvalue_1 = '前一个月个股总市值';
run;
data vwindex2; set vwindex1;
where marketvalue_1^=. ; /*前一个月个股总市值不缺省，则进入本月的总市值加权计算*/
label return = ' ';
run;
proc sort data=vwindex2 out=vwindex3; by year month stkcd; run;
proc means data=vwindex3 noprint;
var return; weight marketvalue_1; /*总市值加权平均求值*/
by year month;
output out=fmreg.vwindex(drop=_freq_ _type_) mean=vwindex;
run;

/*1.3.2排序前贝塔*/
data vwindex4; set fmreg.vwindex; where mdy(month,1,year) between
'1jun1994'd and '1dec2013'd; run;
data vwindex5; set vwindex4; vwindex_1 = lag(vwindex); if vwindex_1 ^= .;
run;

proc sort data=monre_everym6 out=stkret1; by year month stkcd; run;
data stkret2; set stkret1; where mdy(month,1,year) between '1jul1994'd and
'1dec2013'd; run;
data beta1;
merge stkret2 vwindex5;
by year month;
keep stkcd return year month vwindex vwindex_1;
run;
proc sort data=beta1 out=beta2; by stkcd year month; run;

/*计算每个构造组合时点每只股票的排序前贝塔的宏*/
%macro prerankbeta;
%do year=1997 %to 2013;
%let year_3 = %eval(&year-3); /*3年前的年份数*/
/*遴选计算数据*/
data prebeta1; set beta2;
where mdy(month,1,year) between "1jul&year_3"d and "1jun&year"d;
label return=' ';
run;
proc means data=prebeta1 noprint;
var return;
output out=prebeta2(drop=_type_) nmiss=miss;
by stkcd;
run;
data prebeta3; set prebeta2;
nomiss = _freq_ - miss; /*非缺省月度*/
if nomiss>=24; /*必须至少有24个月度非缺省值*/
keep stkcd;
run; /*prebeta3数据集，包含构造组合时点前3年里，至少有24个非缺省月度回报值的股票代码*/

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data prebeta4;
merge prebeta1 prebeta3(in=id);
by stkcd;
if id;
run; /*prebeta4数据集，最终可以进行排序前贝塔计算的所有股票*/

/*回归计算*/
proc reg data=prebeta4 noprint outest=prebeta5;
model return = vindex vindex_1;
by stkcd;
run;
data prebeta6; set prebeta5; keep stkcd year prebeta;
prebeta = vindex + vindex_1; /*sum betas*/
year = &year - 1; /*为了和财务因子一致，此处年份数用前一年表示（即构造组合时点的最近财务年份数）*/
run;

/*合并结果*/
%if &year=1997 %then %do;
data prebeta_all; set prebeta6; delete; run;
%end;
data prebeta_all;
set prebeta_all prebeta6;
run;
%end;
%mend;
%prerankbeta;
/*加市场标识*/
data prebeta_all; set prebeta_all;
if substr(stkcd,1,1)='6' then market=1; /*上海主板*/
if substr(stkcd,1,3)='000' or substr(stkcd,1,3)='001' then market=2; /*深圳主板*/
if substr(stkcd,1,3)='002' then market=3; /*中小板*/
if substr(stkcd,1,1)='3' then market=4; /*创业板*/
run;

/*****步骤2：非正式检测（排序分组）*****/
/*****table1*****/
/*2.1.1指标排序*/
data a1; set Monre_everym6;
where mdy(month,1,year) between '1jun1997'd and '1dec2013'd;
keep stkcd marketvalue year month;
label marketvalue = ' ';
run;
data me; set a1; where month=6; year=year-1; if marketvalue^=.; drop month;
run;
proc sort data=me; by year stkcd; run;
/*为了和财务因子一致，此处年份数用前一年表示（即构造组合时点的最近财务年份数）*/
/*加市场标识*/
data me; set me;
if substr(stkcd,1,1)='6' then market=1; /*上海主板*/
if substr(stkcd,1,3)='000' or substr(stkcd,1,3)='001' then market=2; /*深圳主板*/

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if substr(stkcd,1,3)='002' then market=3; /*中小板*/
if substr(stkcd,1,1)='3' then market=4; /*创业板*/
run;

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data me_prebeta1;
merge me(in=id1) prebeta_all(in=id2 drop=market);
by year stkcd;
if id1 and id2; /*市值和贝塔必须同时存在才能进入分组（此步骤必须在分组前）*/
run;

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data finastkcd; set fmreg.finastkcd; run;
proc sort data=finastkcd; by year stkcd; run;
data me_prebeta2;
merge me_prebeta1 finastkcd(in=id);
by year stkcd;
if id=0; /*去除金融股*/
run;

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/*生成排序宏，并对指标进行排序（以上海主板市场分位数为依据）*/

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%macro rank(infile=,proxy=,outfile=);
    *提取数据;
    data x1; set &infile; where market=1; run;
    proc sort data=x1; by year stkcd; run;
    *计算分位数;
    proc univariate data=x1 noprint;
    var &proxy;
    by year;
    output out=x2 pctlpts = 10 to 90 by 10 pctlpre=dec;
    run;
    *排序;
    proc sql;
    create table &outfile as
    select a.stkcd, a.year, a.&proxy,
    case when missing(a.&proxy) or missing(b.dec10) then .
    when a.&proxy <= b.dec10 then 1
    when a.&proxy <= b.dec20 then 2
    when a.&proxy <= b.dec30 then 3
    when a.&proxy <= b.dec40 then 4
    when a.&proxy <= b.dec50 then 5
    when a.&proxy <= b.dec60 then 6
    when a.&proxy <= b.dec70 then 7
    when a.&proxy <= b.dec80 then 8
    when a.&proxy <= b.dec90 then 9
    when a.&proxy > b.dec90 then 10
    end as r_&proxy
    from &infile as a left join x2 as b
    on a.year=b.year
    order by a.stkcd, a.year, r_&proxy
    ;
quit;

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/*数据集r_x，包含个股每年排序的组序数据，并以代码、年排序*/

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/* proc datasets lib=work nolist; delete x1-x3; run; quit; */
%mend;
%rank(infile=me_prebeta2,proxy=marketvalue,outfile=r_me);
%rank(infile=me_prebeta2,proxy=prebeta,outfile=r_prebeta);

/*市场检测*/
/*
proc format;
value market 1='上海主板'
              2='深圳主板'
              3='中小板'
              4='创业板'
              ;
run;
*检测不同分组的市场分布

proc sort data=aal; by year r_marketvalue stkcd; run;
proc freq data=aal noprint;
table market / out=aa2;
by year r_marketvalue;
format market market.;
run;

proc sort data=aa3; by year r_prebeta stkcd; run;
proc freq data=aa3 noprint;
table market / out=aa4;
by year r_prebeta;
format market market.;
run;
*/

data a2;
merge r_me r_prebeta(drop=prebeta);
by stkcd year;
rename year=financeyear;
run;

/*2.1.2将分组信息和收益信息对应*/
data a3; set Monre_everym6;
where mdy(month,1,year) between '1jul1997'd and '1dec2013'd;
keep stkcd return year month;
label return=' ';
run;
data a4; set a3;
if month >= 7 then financeyear = year - 1; /*7月以后,用的是去年的财务指标*/
else financeyear = year - 2; /*7月之前,用的是前年的财务指标*/
run;
data a5;
merge a4 a2(in=id);
by stkcd financeyear;
if id; /*分组信息必须有,收益在分组信息存在的情况下可以缺省*/
run;
data a6; set a5;
return = return * 100;

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if return=. then return=0;
run;

/*2.1.3计算排序后贝塔*/
proc sql; /*计算size-beta组合平均收益时序*/
create table a7 as select
a2.stkcd, a2.financeyear, a2.r_marketvalue, a2.r_prebeta, a3.year,
a3.month, a3.return
from a3, a2
where a3.stkcd=a2.stkcd
order by a2.financeyear, a2.r_marketvalue, a2.r_prebeta, a3.year, a3.month,
a2.stkcd
;
quit;

proc means data=a7 noprint;
var return;
by financeyear r_marketvalue r_prebeta year month;
output out=a8(drop=_freq_ _type_) mean=ewret;
run;
proc sql; /*计算postbeta*/
create table a9 as select
a8.*, vwinindex5.vwinindex, vwinindex5.vwinindex_1
from a8, vwinindex5
where a8.year=vwinindex5.year and a8.month=vwinindex5.month
order by a8.financeyear, a8.r_marketvalue, a8.r_prebeta, a8.year, a8.month
;
quit;

proc reg data=a9 outest=a10 noprint;
model ewret = vwinindex vwinindex_1;
by financeyear r_marketvalue r_prebeta;
run;

data a11; set a10; keep financeyear r_marketvalue r_prebeta postbeta;
postbeta = vwinindex + vwinindex_1; /*sum betas*/
rename financeyear=year;
run;

proc sql; /*将组合收益时序的postbeta赋予单只股票*/
create table postbeta_all as select
a2.stkcd, a11.year, a11.postbeta
from a2, a11
where a2.financeyear=a11.year and a2.r_marketvalue=a11.r_marketvalue and
a2.r_prebeta=a11.r_prebeta
order by a11.year, a2.stkcd
;
quit;

/*2.1.4计算分组值*/
proc sql;
create table a12 as select
a6.financeyear, a6.stkcd, a6.year, a6.month, a6.r_marketvalue,
a6.r_prebeta, a6.return, a6.marketvalue, b.postbeta
from a6, postbeta_all as b

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where a6.financeyear=b.year and a6.stkcd=b.stkcd
order by a6.year, a6.month, a6.r_marketvalue, a6.r_prebeta, a6.stkcd
;
quit;
data a13; set a12; me = log(marketvalue/1000000); run; /*a13数据集, 下面各
类计算的输入集*/

/*按照贝塔、市值分组*/
proc means data=a13 noprint; /*单个月度内的股票进行平均*/
var return me postbeta;
by year month r_marketvalue r_prebeta;
output out=a14(drop=_freq_ _type_) mean=portret portme portbeta;
run;
proc sort data=a14 out=a15; by r_marketvalue r_prebeta; run; /*size-beta
组合的所有月度值进行平均*/
proc means data=a15 noprint;
var portret portme portbeta;
by r_marketvalue r_prebeta;
output out=a16(drop=_freq_ _type_) mean=portret portme portbeta;
run;

/*仅按照总市值分组*/
proc means data=a13 noprint; /*单个月度内的股票进行平均*/
var return me postbeta;
by year month r_marketvalue;
output out=a17(drop=_freq_ _type_) mean=portret portme portbeta;
run;
proc sort data=a17 out=a18; by r_marketvalue; run; /*size-beta组合的所有月
度值进行平均*/
proc means data=a18 noprint;
var portret portme portbeta;
by r_marketvalue;
output out=a19(drop=_freq_ _type_) mean=portret portme portbeta;
run;

/*仅按照贝塔分组*/
proc sort data=a13 out=a20; by r_prebeta year month stkcd; run;
proc means data=a20 noprint; /*单个月度内的股票进行平均*/
var return me postbeta;
by r_prebeta year month;
output out=a21(drop=_freq_ _type_) mean=portret portme portbeta;
run;
proc means data=a21 noprint; /*size-beta组合的所有月度值进行平均*/
var portret portme portbeta;
by r_prebeta;
output out=a22(drop=_freq_ _type_) mean=portret portme portbeta;
run;

/*无分组*/
proc means data=a13 noprint; /*单个月度内的股票进行平均*/
var return me postbeta;
by year month;
output out=a23(drop=_freq_ _type_) mean=portret portme portbeta;
run;

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proc means data=a23 noprint; /*size-beta组合的所有月度值进行平均*/
var portret portme portbeta;
output out=a24(drop=_freq_ _type_) mean=portret portme portbeta;
run;

%macro table1(outfile=, var=);
    *两值分组;
    data x1; set a16; keep r_marketvalue r_prebeta &var; run;
    proc transpose data=x1 out=x2 prefix=beta;
    var &var;
    by r_marketvalue;
    id r_prebeta;
    run;
    data x3; set x2;
    rename _name_ = table1; label _name_ = ' ';
    _name_ = cats('ME', r_marketvalue);
    run;
    *市值分组;
    data x4; set a19; keep r_marketvalue &var; rename &var=all; run;
    *贝塔分组;
    proc transpose data=a22 out=x5 prefix=beta;
    var &var;
    id r_prebeta;
    run;
    data x6; set x5; drop _name_; run;
    *无分组;
    data x7; set a24; keep &var; rename &var=all; run;
    *合并结果;
    data x8;
    merge x4 x3;
    by r_marketvalue;
    drop r_marketvalue;
    run;
    proc sql; create table x9 as select 'all' as table1 length 8, x7.*, x6.*
from x7, x6; quit;
    data &outfile; set x9 x8; run;
%mend;

%table1(outfile=table1a, var=portret);
%table1(outfile=table1b, var=portbeta);
%table1(outfile=table1c, var=portme);

* table1 print;
    title "Table 1a";
    title2 "size-beta组合的月度平均收益 (1997/7-2013/12) ";
    proc print data=table1a;
        format _numeric_ comma5.2;
    run;

    title "Table 1b";
    title2 "size-beta组合的月度平均排序后beta (1997/7-2013/12) ";
    proc print data=table1b;
        format _numeric_ comma5.2;

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run;

title "Table 1c";
title2 "size-beta组合的月度平均市值（1997/7-2013/12）";
proc print data=table1c;
    format _numeric_ comma5.2;
run;

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/*****table2*****/
/*2.2.1对市值或者贝塔进行排序*/
/*生成排序宏，并对指标进行排序（以上海主板市场分位数为依据，分位数变为5,10-90,95）*/
data b1; set me_prebeta2; run; /*用me_prebeta2作为输入集，使table2和table1
一致*/
%macro rank_single(infile=,proxy=,outfile=);
    *提取数据;
    data x1; set &infile; where market=1; run;
    proc sort data=x1; by year stkcd; run;
    *计算分位数;
    proc univariate data=x1 noprint;
    var &proxy;
    by year;
    output out=x2 pctlpts = 5 10 20 30 40 50 60 70 80 90 95 pctlpre=dec;
    run;
    *排序;
    proc sql;
    create table &outfile as
    select a.stkcd, a.market, a.year, a.&proxy,
    case when missing(a.&proxy) or missing(b.dec10) then .
    when a.&proxy <= b.dec5 then 1
    when a.&proxy <= b.dec10 then 2
    when a.&proxy <= b.dec20 then 3
    when a.&proxy <= b.dec30 then 4
    when a.&proxy <= b.dec40 then 5
    when a.&proxy <= b.dec50 then 6
    when a.&proxy <= b.dec60 then 7
    when a.&proxy <= b.dec70 then 8
    when a.&proxy <= b.dec80 then 9
    when a.&proxy <= b.dec90 then 10
    when a.&proxy <= b.dec95 then 11
    when a.&proxy > b.dec95 then 12
    end as r_&proxy
    from &infile as a left join x2 as b
    on a.year=b.year
    order by a.stkcd, a.year, r_&proxy
    ;
quit;

/*数据集r_x，包含个股每年排序的组序数据，并以代码、年排序*/

```

```

/* proc datasets lib=work nolist; delete x1-x3; run; quit; */
%mend;
%rank_single(infile=b1,proxy=marketvalue,outfile=r_me_single);
%rank_single(infile=b1,proxy=prebeta,outfile=r_prebeta_single);
data b2;
merge r_me_single r_prebeta_single;
by stkcd year;
drop prebeta;
run;

/*合并数据*/
proc sort data=b2 out=b3; by year stkcd r_marketvalue; run;
%macro merge(base=,merge=,outfile=,var=,ifmiss=0);
*ifmiss=1时, 表示合并指标的缺省值会被一个新增的虚拟变量表示(检测用);
data x1;
merge &base(in=id) &merge;
by year stkcd;
if id;
run;
data &outfile; set x1;
%if &ifmiss=1 %then %do; if &var = . then &var._miss = 1; else &var._miss
= 0; %end;
run;
%mend;
%merge(base=b3,merge=bm,outfile=b4,var=bm);
%merge(base=b4,merge=am,outfile=b5,var=am);
%merge(base=b5,merge=ab,outfile=b6,var=ab);
%merge(base=b6,merge=ep_positive,outfile=b7,var=ep_positive);
%merge(base=b7,merge=ep_negative,outfile=b8,var=ep_negative);
%merge(base=b8,merge=rm,outfile=b9,var=rm);
%merge(base=b9,merge=postbeta_all,outfile=b10,var=postbeta);
proc sql;
create table b11 as select
stkcd, market, year, r_marketvalue, r_prebeta, postbeta,
(marketvalue/1000000) as me,
bm, am, ab, ep_positive, ep_negative, rm
from b10
;
quit;
data b12; set b11;
me = log(me); bm = log(bm); am = log(am); ab = log(ab); rm = log(rm);
rename year = financeyear;
run;

/*将分组、指标信息和收益信息对应*/
data b13; set Monre_everym6;
where mdy(month,1,year) between '1jul1997'd and '1dec2013'd;
keep stkcd return year month;
label return=' ';
run;
data b14; set b13;
if month >= 7 then financeyear = year - 1; /*7月以后,用的是去年的财务指标*/
else financeyear = year - 2; /*7月之前,用的是前年的财务指标*/
run;

proc sort data=b12; by stkcd financeyear; run;

```

```

data b15;
merge b14 b12(in=id);
by stkcd financeyear;
if id; /*分组信息必须有，收益在分组信息存在的情况下可以缺省*/
run;
data b16; set b15;
return = return * 100;
if return=. then return=0;
run;

/*回归数据集fmreg.fmregression*/
data fmreg.fmregression; set b16; drop r_marketvalue r_prebeta; run;
proc sort data=fmreg.fmregression; by year month stkcd; run;

/*2.2.2计算每组的各个指标均值*/
/*市值排序*/
proc sort data=b16 out=b17; by r_marketvalue year month stkcd; run;
proc means data=b17 noprint; /*单个月度内的股票进行平均*/
var return me postbeta bm am ab ep_positive ep_negative rm;
by r_marketvalue year month;
output out=b18(drop=_freq_ _type_) mean=return me postbeta bm am ab
ep_positive ep_negative rm;
run;
proc means data=b18 noprint; /*size-beta组合的所有月度值进行平均*/
var return me postbeta bm am ab ep_positive ep_negative rm;
by r_marketvalue;
output out=b19(drop=_freq_ _type_) mean=return me postbeta bm am ab
ep_positive ep_negative rm;
run;
proc means data=b19 noprint; /*平均公司家数*/
var r_marketvalue;
by r_marketvalue;
output out=b20(drop=_freq_ _type_) n=firms;
run;
proc means data=b20 noprint;
var firms;
by r_marketvalue;
output out=b21(drop=_freq_ _type_) mean=firms;
run;
data b22; /*合并结果*/
merge b19 b21; by r_marketvalue;
run;
proc transpose data=b22 out=b23 prefix=ME;
id r_marketvalue;
run;
data table2a; set b23; label _name_ = 'table2'; run;
* table2 print;
title "Table 2a";
title2 "size组合的特征 (1997/7-2013/12) ";
proc print data=table2a;
format _numeric_ comma8.2;
run;

/*贝塔排序*/

```

```

proc sort data=b16 out=b24; by r_prebeta year month stkcd; run;
proc means data=b24 noprint; /*单个月度内的股票进行平均*/
var return me postbeta bm am ab ep_positive ep_negative rm;
by r_prebeta year month;
output out=b25(drop=_freq_ _type_) mean=return me postbeta bm am ab
ep_positive ep_negative rm;
run;

proc means data=b25 noprint; /*size-beta组合的所有月度值进行平均*/
var return me postbeta bm am ab ep_positive ep_negative rm;
by r_prebeta;
output out=b26(drop=_freq_ _type_) mean=return me postbeta bm am ab
ep_positive ep_negative rm;;
run;

proc means data=b24 noprint; /*平均公司家数*/
var r_prebeta;
by r_prebeta ;
output out=b27(drop=_freq_ _type_) n=firms;
run;
proc means data=b27 noprint;
var firms;
by r_prebeta;
output out=b28(drop=_freq_ _type_) mean=firms;
run;

data b29; /*合并结果*/
merge b26 b28; by r_prebeta;
run;
proc transpose data=b29 out=b30 prefix=beta;
id r_prebeta;
run;
data table2b; set b30; label _name_ = 'table2'; run;
* table2 print;
title "Table 2b";
title2 "beta组合的特征 (1997/7-2013/12) ";
proc print data=table2b;
format _numeric_ comma8.2;
run;

/*****步骤3: FM回归*****/
/*****table3*****/
data c1; set fmreg.fmregression; run;
/*3.1极端值缩尾处理*/
%macro winsorize(infile=,outfile=,statistic=,var=,ifwinsorize=1);
*描述性统计;
proc univariate data=&infile noprint;
var &var;
output out=&statistic
N=num nmiss=miss min=min pctlpts=0.5 25 50 75 99.5 pctlpre=per max=max;
run;
*缩尾处理;
%if &ifwinsorize=1 %then %do;
proc sql;
create table x1 as select
a.*, b.per0_5, b.per99_5
from &infile as a, &statistic as b
;

```

```

quit;
data &outfile; set x1;
if &var >= per99_5 then &var = per99_5;
if &var <= per0_5 then &var = per0_5;
drop per0_5 per99_5;
run;
%end;
%else %do;
data &outfile; set &infile; run;
%end;
%mend;
%winsorize(infile=c1,outfile=c2,statistic=c2_stat,var=bm,ifwinsorize=1)
;
%winsorize(infile=c2,outfile=c3,statistic=c3_stat,var=am,ifwinsorize=1)
;
%winsorize(infile=c3,outfile=c4,statistic=c4_stat,var=ab,ifwinsorize=1)
;
%winsorize(infile=c4,outfile=c5,statistic=c5_stat,var=ep_positive,ifwinsorize=1);
%winsorize(infile=c5,outfile=c6,statistic=c6_stat,var=rm,ifwinsorize=1)
;

/*3.2进行FM回归*/
/*构造一个FM回归宏*/
%macro FMreg(infile=,outfile=,var1=,var2=,var3=,var4=,var5=);
    *每月截面回归;
    proc reg data=&infile outest=x1 noprint;
    model return = &var1 &var2 &var3 &var4 &var5;
    by year month;
    run;
    data x2; set x1;
    keep year month &var1 &var2 &var3 &var4 &var5;
    run;
    *各个月求均值和t值;
    proc means data=x2 noprint;
    var &var1 &var2 &var3 &var4 &var5;
    output out=x3(drop=_freq_ _type_) mean=;
    run;
    data x4;
    informat item $12.; format item $12.;
    set x3;
    item = 'Mean';
    run;
    proc means data=x2 noprint;
    var &var1 &var2 &var3 &var4 &var5;
    output out=x5(drop=_freq_ _type_) t=;
    run;
    data x6;
    informat item $12.; format item $12.;
    set x5;
    item = 't-statistic';
    run;
    data &outfile; set x4 x6; run;
%mend;
%FMreg(infile=c6,outfile=ret_beta,var1=postbeta);

```

```

%FMreg(infile=c6,outfile=ret_me,var1=me);
%FMreg(infile=c6,outfile=ret_beta_me,var1=postbeta,var2=me);
%FMreg(infile=c6,outfile=ret_bm,var1=bm);
%FMreg(infile=c6,outfile=ret_am_ab,var1=am,var2=ab);
%FMreg(infile=c6,outfile=ret_epn_epp,var1=ep_negative,var2=ep_positive)
;
%FMreg(infile=c6,outfile=ret_me_bm,var1=me,var2=bm);
%FMreg(infile=c6,outfile=ret_me_am_ab,var1=me,var2=am,var3=ab);
%FMreg(infile=c6,outfile=ret_me_epn_epp,var1=me,var2=ep_negative,var3=ep_positive);
%FMreg(infile=c6,outfile=ret_me_bm_epn_epp,var1=me,var2=bm,var3=ep_negative,var4=ep_positive);
%FMreg(infile=c6,outfile=ret_me_am_ab_epn_epp,var1=me,var2=am,var3=ab,var4=ep_negative,var5=ep_positive);

*添加指标RM;
%FMreg(infile=c6,outfile=ret_rm,var1=rm);
%FMreg(infile=c6,outfile=ret_me_rm,var1=me,var2=rm);
%FMreg(infile=c6,outfile=ret_bm_rm,var1=bm,var2=rm);
%FMreg(infile=c6,outfile=ret_epn_epp_rm,var1=rm,var2=ep_negative,var3=ep_positive);
%FMreg(infile=c6,outfile=ret_me_epn_epp_rm,var1=me,var2=ep_negative,var3=ep_positive,var4=rm);
%FMreg(infile=c6,outfile=ret_me_bm_rm,var1=me,var2=bm,var3=rm);

/*
检验贝塔平方项
data c7; set c6;
postbeta2 = (postbeta) ** 2;
run;
%FMreg(infile=c7,outfile=ret_beta2,var1=postbeta2);
%FMreg(infile=c7,outfile=ret_beta_beta2,var1=postbeta,var2=postbeta2);
*/

data table3a;
set ret_beta ret_beta_me ret_bm ret_am_ab ret_epn_epp ret_me_bm ret_me_am_ab
ret_me_epn_epp ret_me_bm_epn_epp ret_me_am_ab_epn_epp ret_rm ret_me_rm
ret_bm_rm ret_epn_epp_rm ret_me_epn_epp_rm ret_me_bm_rm
;
run;
data fmreg.result_all; set table3a; run;
* table3 print;
title "Table 3a";
title2 "全市场的FM回归结果 (1997/7-2013/12) ";
proc print data=table3a;
format _numeric_ comma8.2;
run;

/*****步骤4：有效变量分组 (a) *****/
/*****table4*****/
/*4.1对市值和市销率进行分组*/
data d1;
merge me(in=id1) rm(in=id2);

```

```

by year stkcd;
if id1 and id2; /*ME和RM必须有值; 注意到这里和table1-2的基础数据有不一致的地方,
前面是ME和prebeta必须有值*/
run;
%rank(infile=d1,proxy=marketvalue,outfile=r_me);
%rank(infile=d1,proxy=rm,outfile=r_rm);
data d2;
merge r_me(keep=year stkcd r_marketvalue) r_rm(keep=year stkcd r_rm);
by stkcd year;
rename year=financeyear;
run;

/*4.2将分组信息和收益信息对应*/
data d3; set Monre_everym6;
where mdy(month,1,year) between '1jul1997'd and '1dec2013'd;
keep stkcd return year month;
label return=' ';
run;
data d4; set d3;
if month >= 7 then financeyear = year - 1; /*7月以后,用的是去年的财务指标*/
else financeyear = year - 2; /*7月之前,用的是前年的财务指标*/
run;
data d5;
merge d4 d2(in=id);
by stkcd financeyear;
if id; /*分组信息必须有,收益在分组信息存在的情况下可以缺省*/
run;
data d6; set d5;
return = return * 100;
if return=. then return=0;
run;

/*4.3各组内求平均值*/
/*两值分组*/
proc sort data=d6; by r_marketvalue r_rm year month stkcd; run;
proc means data=d6 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue r_rm year month;
output out=d7(drop=_freq_ _type_) mean=portret;
run;
proc means data=d7 noprint; /*size-rm组合的所有月度值进行平均*/
var portret;
by r_marketvalue r_rm;
output out=d8(drop=_freq_ _type_) mean=portret;
run;

/*单总市值分组*/
proc sort data=d6 out=d9; by r_marketvalue year month; run;
proc means data=d9 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue year month;
output out=d10(drop=_freq_ _type_) mean=portret;
run;

```

```

proc means data=d10 noprint; /*size-rm组合的所有月度值进行平均*/
var portret;
by r_marketvalue;
output out=d11(drop=_freq_ _type_) mean=portret;
run;

/*单市销率分组*/
proc sort data=d6 out=d12; by r_rm year month; run;
proc means data=d12 noprint; /*单个月度内的股票进行平均*/
var return;
by r_rm year month;
output out=d13(drop=_freq_ _type_) mean=portret;
run;
proc means data=d13 noprint; /*size-rm组合的所有月度值进行平均*/
var portret;
by r_rm;
output out=d14(drop=_freq_ _type_) mean=portret;
run;

/*无分组*/
proc sort data=d6 out=d15; by year month; run;
proc means data=d15 noprint; /*单个月度内的股票进行平均*/
var return;
by year month;
output out=d16(drop=_freq_ _type_) mean=portret;
run;
proc means data=d16 noprint; /*size-rm组合的所有月度值进行平均*/
var portret;
output out=d17(drop=_freq_ _type_) mean=portret;
run;

%macro table4(outfile=, var=);
    *两值分组;
    data x1; set d8; keep r_marketvalue r_rm &var; run;
    proc transpose data=x1 out=x2 prefix=rm;
    var &var;
    by r_marketvalue;
    id r_rm;
    run;
    data x3; set x2;
    rename _name_ = table4; label _name_ = ' ';
    _name_ = cats('ME', r_marketvalue);
    run;
    *市值分组;
    data x4; set d11; keep r_marketvalue &var; rename &var=all; run;
    *市销率分组;
    proc transpose data=d14 out=x5 prefix=rm;
    var &var;
    id r_rm;
    run;
    data x6; set x5; drop _name_; run;
    *无分组;
    data x7; set d17; keep &var; rename &var=all; run;
    *合并结果;

```

```

data x8;
merge x4 x3;
by r_marketvalue;
drop r_marketvalue;
run;
proc sql; create table x9 as select 'all' as table4 length 8, x7.*, x6.*
from x7, x6; quit;
data &outfile; set x9 x8; run;
%mend;
%table4(outfile=table4, var=portret);
* table4 print;
title "Table 4";
title2 "size-RM组合的月度平均收益 (1997/7-2013/12) ";
proc print data=table4;
format _numeric_ comma8.2;
run;

```

```

/*****步骤5: 有效变量分组 (b) *****/
/*****table5*****/
/*5.1对市值和市净率进行分组*/
data e1;
merge me(in=id1) bm(in=id2);
by year stkcd;
if id1 and id2; /*ME和BM必须有值; 注意到这里和table1-2的基础数据有不一致的地方,
前面是ME和prebeta必须有值*/
run;
%rank(infile=e1,proxy=marketvalue,outfile=r_me);
%rank(infile=e1,proxy=bm,outfile=r_bm);
data e2;
merge r_me(keep=year stkcd r_marketvalue) r_bm(keep=year stkcd r_bm);
by stkcd year;
rename year=financeyear;
run;

/*5.2将分组信息和收益信息对应*/
data e3; set Monre_everym6;
where mdy(month,1,year) between '1jul1997'd and '1dec2013'd;
keep stkcd return year month;
label return=' ';
run;
data e4; set e3;
if month >= 7 then financeyear = year - 1; /*7月以后, 用的是去年的财务指标*/
else financeyear = year - 2; /*7月之前, 用的是前年的财务指标*/
run;
data e5;
merge e4 e2(in=id);
by stkcd financeyear;
if id; /*分组信息必须有, 收益在分组信息存在的情况下可以缺省*/
run;
data e6; set e5;
return = return * 100;

```

```

if return=. then return=0;
run;

/*5.3各组内求平均值*/
/*两值分组*/
proc sort data=e6; by r_marketvalue r_bm year month stkcd; run;
proc means data=e6 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue r_bm year month;
output out=e7(drop=_freq_ _type_) mean=portret;
run;

proc means data=e7 noprint; /*size-bm组合的所有月度值进行平均*/
var portret;
by r_marketvalue r_bm;
output out=e8(drop=_freq_ _type_) mean=portret;
run;

/*单总市值分组*/
proc sort data=e6 out=e9; by r_marketvalue year month; run;
proc means data=e9 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue year month;
output out=e10(drop=_freq_ _type_) mean=portret;
run;

proc means data=e10 noprint; /*size-bm组合的所有月度值进行平均*/
var portret;
by r_marketvalue;
output out=e11(drop=_freq_ _type_) mean=portret;
run;

/*单市净率分组*/
proc sort data=e6 out=e12; by r_bm year month; run;
proc means data=e12 noprint; /*单个月度内的股票进行平均*/
var return;
by r_bm year month;
output out=e13(drop=_freq_ _type_) mean=portret;
run;

proc means data=e13 noprint; /*size-bm组合的所有月度值进行平均*/
var portret;
by r_bm;
output out=e14(drop=_freq_ _type_) mean=portret;
run;

/*无分组*/
proc sort data=e6 out=e15; by year month; run;
proc means data=e15 noprint; /*单个月度内的股票进行平均*/
var return;
by year month;
output out=e16(drop=_freq_ _type_) mean=portret;
run;

proc means data=e16 noprint; /*size-bm组合的所有月度值进行平均*/
var portret;
output out=e17(drop=_freq_ _type_) mean=portret;
run;

```

```

%macro table5(outfile=, var=);
    *两值分组;
    data x1; set e8; keep r_marketvalue r_bm &var; run;
    proc transpose data=x1 out=x2 prefix=bm;
    var &var;
    by r_marketvalue;
    id r_bm;
    run;
    data x3; set x2;
    rename _name_ = table5; label _name_ = ' ';
    _name_ = cats('ME', r_marketvalue);
    run;
    *市值分组;
    data x4; set e11; keep r_marketvalue &var; rename &var=all; run;
    *市净率分组;
    proc transpose data=e14 out=x5 prefix=bm;
    var &var;
    id r_bm;
    run;
    data x6; set x5; drop _name_; run;
    *无分组;
    data x7; set e17; keep &var; rename &var=all; run;
    *合并结果;
    data x8;
    merge x4 x3;
    by r_marketvalue;
    drop r_marketvalue;
    run;
    proc sql; create table x9 as select 'all' as table5 length 8, x7.*, x6.*
from x7, x6; quit;
    data &outfile; set x9 x8; run;
%mend;
%table5(outfile=table5, var=portret);
* table5 print;
    title "Table 5";
    title2 "size-BM组合的月度平均收益 (1997/7-2013/12) ";
    proc print data=table5;
        format _numeric_ comma8.2;
    run;

```

```

/*****步骤6: 有效变量分组 (c) *****/
/*****table6*****/
/*6.1对市值和市盈率进行分组*/
*市盈率特殊之处在于亏损时分组标识为0;
data f1;
merge me(in=id1) ep_all(in=id2);
by year stkcd;
if id1 and id2; /*ME和EP必须有值; 注意到这里和table1-2的基础数据有不一致的地方,
前面是ME和prebeta必须有值*/
run;

```

```

%rank(infile=f1,proxy=marketvalue,outfile=r_me);
data f2; set f1; where ep >= 0; run;
%rank(infile=f2,proxy=ep,outfile=r_ep);
data f3;
merge r_me(keep=year stkcd r_marketvalue in=id1) r_ep(keep=year stkcd r_ep
in=id2);
by stkcd year;
rename year=financyear;
if id1=1 and id2=0 then r_ep=0;
run;

```

/*6.2将分组信息和收益信息对应*/

```

data f4; set Monre_everym6;
where mdy(month,1,year) between '1jul1997'd and '1dec2013'd;
keep stkcd return year month;
label return=' ';
run;
data f5; set f4;
if month >= 7 then financyear = year - 1; /*7月以后，用的是去年的财务指标*/
else financyear = year - 2; /*7月之前，用的是前年的财务指标*/
run;
data f6;
merge f5 f3(in=id);
by stkcd financyear;
if id; /*分组信息必须有，收益在分组信息存在的情况下可以缺省*/
run;
data f7; set f6;
return = return * 100;
if return=. then return=0;
run;

```

/*6.3各组内求平均值*/

/*两值分组*/

```

proc sort data=f7; by r_marketvalue r_ep year month stkcd; run;
proc means data=f7 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue r_ep year month;
output out=f8(drop=_freq_ _type_) mean=portret;
run;
proc means data=f8 noprint; /*size-ep组合的所有月度值进行平均*/
var portret;
by r_marketvalue r_ep;
output out=f9(drop=_freq_ _type_) mean=portret;
run;

```

/*单总市值分组*/

```

proc sort data=f7 out=f10; by r_marketvalue year month; run;
proc means data=f10 noprint; /*单个月度内的股票进行平均*/
var return;
by r_marketvalue year month;
output out=f11(drop=_freq_ _type_) mean=portret;
run;
proc means data=f11 noprint; /*size-ep组合的所有月度值进行平均*/

```

```

var portret;
by r_marketvalue;
output out=f12(drop=_freq_ _type_) mean=portret;
run;

/*单市盈率分组*/
proc sort data=f7 out=f13; by r_ep year month; run;
proc means data=f13 noprint; /*单个月度内的股票进行平均*/
var return;
by r_ep year month;
output out=f14(drop=_freq_ _type_) mean=portret;
run;
proc means data=f14 noprint; /*size-ep组合的所有月度值进行平均*/
var portret;
by r_ep;
output out=f15(drop=_freq_ _type_) mean=portret;
run;

/*无分组*/
proc sort data=f7 out=f16; by year month; run;
proc means data=f16 noprint; /*单个月度内的股票进行平均*/
var return;
by year month;
output out=f17(drop=_freq_ _type_) mean=portret;
run;
proc means data=f17 noprint; /*size-bm组合的所有月度值进行平均*/
var portret;
output out=f18(drop=_freq_ _type_) mean=portret;
run;

%macro table6(outfile=, var=);
    *两值分组;
    data x1; set f9; keep r_marketvalue r_ep &var; run;
    proc transpose data=x1 out=x2 prefix=ep;
    var &var;
    by r_marketvalue;
    id r_ep;
    run;
    data x3; set x2;
    rename _name_ = table6; label _name_ = ' ';
    _name_ = cats('ME', r_marketvalue);
    run;
    *市值分组;
    data x4; set f12; keep r_marketvalue &var; rename &var=all; run;
    *市盈率分组;
    proc transpose data=f15 out=x5 prefix=ep;
    var &var;
    id r_ep;
    run;
    data x6; set x5; drop _name_; run;
    *无分组;
    data x7; set f18; keep &var; rename &var=all; run;
    *合并结果;

```

```

data x8;
merge x4 x3;
by r_marketvalue;
drop r_marketvalue;
run;
proc sql; create table x9 as select 'all' as table6 length 8, x7.*, x6.*
from x7, x6; quit;
data &outfile; set x9 x8; run;
%mend;
%table6(outfile=table6, var=portret);
* table6 print;
title "Table 6";
title2 "size-EP组合的月度平均收益（1997/7-2013/12）";
footnote "ep0是所有亏损公司分组";
proc print data=table6;
format _numeric_ comma8.2;
run;

ods html close;
ods html;

```